Northwestern at DialectSentEval 2026: A Comparative Study of MARBERTv2 and AraBERT for Arabic Dialect Sentiment Analysis

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Abstract

This paper describes our submission to **Subtask 1 of DialectSentEval 2026**, which focuses on three class sentiment classification for Arabic dialectal text. The task aims to classify sentences written in five Arabic dialects into *positive*, *negative*, or *neutral* sentiment categories. We investigate the effectiveness of transformer-based Arabic language models through a series of controlled experiments. Specifically, we compare MARBERTv2 and AraBERT while maintaining identical pre processing, training, and evaluation settings to ensure a fair comparison. In addition, we examine whether explicitly providing dialect information as part of the input improves classification performance.

Experimental results on an internal validation split show that MARBERTv2 consistently outperforms AraBERT on this task. Furthermore, removing the explicit dialect tag from the input slightly improves performance, suggesting that MARBERTv2 can effectively capture dialect-specific linguistic patterns directly from the sentence. Our best configuration, which uses MARBERTv2 with sentence-only input, achieves a Macro F1-score of **0.9394** and an accuracy of **94.24%**. These findings demonstrate the effectiveness of dialect-aware pre-trained language models for Arabic sentiment analysis and establish a strong baseline for the official shared task evaluation.

1 Introduction

Sentiment analysis is a fundamental Natural Language Processing (NLP) task that aims to identify the polarity of opinions expressed in text. Despite the remarkable progress of transformer-based language models, Arabic dialect sentiment analysis remains challenging due to substantial lexical, morphological, and syntactic variation across regional dialects.

The DialectSentEval 2026 Shared Task (Ezzini et al., 2026a) (Ezzini et al., 2026b) addresses this challenge by introducing a benchmark for sentiment classification across five Arabic dialects: Moroccan, Egyptian, Lebanese, Jordanian, and Saudi. In Subtask 1, systems are required to classify each input sentence into one of three sentiment categories: positive, negative, or neutral.To address this task, we conduct a controlled empirical comparison of two Arabic pretrained transformer models, MARBERTv2 and AraBERT. In addition, we investigate whether explicitly incorporating dialect information into the model input improves sentiment classification performance. All experiments employ identical preprocessing, data splitting, training, and evaluation settings to ensure a fair comparison between model architectures and input representations.Experimental results on an internal validation split show that MARBERTv2 consistently outperforms AraBERT. Furthermore, explicitly prepending dialect labels does not improve performance, indicating that MARBERTv2 effectively captures dialect specific linguistic characteristics directly from the input sentence.The best performing configuration achieves a Macro F1-score of 0.9394 using sentence only input, demonstrating the effectiveness of dialect aware pre trained language models for Arabic sentiment analysis.

Recent work has further evaluated large language models for Arabic dialect sentiment analysis and introduced shared-task benchmarks for sentiment analysis across Arabic dialects (Alharbi et al., 2025a,b).

2 Background

DialectSentEval 2026 Subtask 1 addresses multiclass sentiment classification for Arabic dialectal text. Given an input sentence written in one of five Arabic dialects, namely Moroccan, Egyptian, Lebanese, Jordanian, and Saudi, the task requires predicting one of three sentiment labels: *positive*,

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1409–1413
October 28-29, 2026 ©2026 Association for Computational Linguistics

negative, or *neutral*. For example, a sentence expressing satisfaction is assigned the label *positive*, whereas a sentence expressing dissatisfaction is labeled *negative*.

The released training dataset contains approximately 1.7K manually annotated Arabic sentences translated from hotel reviews and annotated by native speakers of the target dialects. During system development, only the released training data were available, while the official test labels remained hidden. Our submission participated in Subtask 1, which focuses on three-class sentiment classification.

Transformer-based pretrained language models have substantially advanced Arabic NLP. AraBERT (Antoun et al., 2020) introduced an Arabic-specific BERT model that achieved state-of-the-art performance on multiple Arabic NLP tasks. MARBERT (Abdul-Mageed et al., 2021) was pretrained on large-scale multi-dialect Arabic Twitter data, while MARBERTv2 further extends MARBERT with additional Modern Standard Arabic data and a longer maximum sequence length. Shared tasks such as NADI (Abdul-Mageed et al., 2022) further highlighted the importance of robust dialect processing, whereas recent surveys (Shi and Agrawal, 2025) identify dialectal diversity and limited annotated resources as major challenges for Arabic sentiment analysis. MA'AKS (Mughaus et al., 2026) introduced a manually curated parallel Arabic dataset for sentiment style transfer, providing 5,000 positive negative sentence pairs.Motivated by these challenges, we conduct a controlled comparison of AraBERT and MARBERTv2 while investigating the impact of explicitly incorporating dialect information to identify the best-performing system for the DialectSentEval 2026 shared task.

3 System Overview

Our system formulates DialectSentEval 2026 Subtask 1 as a supervised three-class sentiment classification task using pretrained transformer-based transformer models. Given an input sentence, the selected tokenizer converts the text into subword tokens, which are processed by the transformer encoder. The contextual representation produced by the encoder is passed to a task-specific classification head that predicts one of three sentiment labels: positive, negative, or neutral.

The proposed system relies on publicly available pretrained language models as external resources beyond the provided training data. Specifically, we evaluate MARBERTv2 and AraBERT, both initialized from their released pretrained checkpoints. No additional datasets, sentiment lexicons, or external linguistic resources are used during training.

To address the lexical and linguistic diversity of Arabic dialects, we investigate two design choices. First, we compare a dialect-oriented pretrained model (MARBERTv2) with a general Arabic pretrained model (AraBERT). Second, we examine whether explicitly providing the dialect name as part of the input improves sentiment classification performance.

Three controlled system configurations are evaluated. The first fine-tunes MARBERTv2 using the sentence preceded by its dialect label (e.g., [Egyptian]). The second fine-tunes MARBERTv2 using only the original sentence without the dialect tag. The third replaces MARBERTv2 with AraBERT while keeping all preprocessing, training, and evaluation settings identical. This controlled design isolates the effect of both the pretrained language model and the input representation on sentiment classification performance.

4 Experimental Setup

Our system is based on transformer-based Arabic language models fine-tuned for three-class sentiment classification. The overall pipeline consists of four main stages: data preparation, text preprocessing, model fine-tuning, and prediction.

Given an input sentence, the text is first preprocessed using the tokenizer associated with the selected pre-trained language model. The tokenized sequence is then passed to a transformer encoder followed by a classification head that predicts one of three sentiment labels: *positive*, *negative*, or *neutral*.To identify the most suitable architecture for the shared task, we conducted three controlled experiments. The first experiment fine-tuned MARBERTv2 using the sentence together with its dialect label as input. The second experiment evaluated MARBERTv2 using only the sentence text, without explicitly providing dialect information. Finally, the third experiment replaced MARBERTv2 with AraBERT while keeping all preprocessing steps, training settings, and evaluation procedures identical. This controlled design allowed us to isolate the effect of the language

model and the input representation on sentiment classification performance. The following subsections describe the dataset, preprocessing pipeline, evaluated models, and experimental setup in detail.

4.1 Dataset

We used the training data provided by the organizers of DialectSentEval 2026 Subtask 1. The dataset consists of approximately 1.7K manually annotated Arabic sentences covering five dialects: Moroccan (Darija), Egyptian, Lebanese, Jordanian, and Saudi. Each sentence is assigned one of three sentiment labels: *positive*, *negative*, or *neutral*.

The dataset originates from hotel reviews that were translated and annotated by native speakers of the corresponding dialects. Since the official test labels were not available during the development phase, all experiments were conducted using the released training data. An internal validation split was created to evaluate different system configurations before the official evaluation phase.

4.2 Preprocessing

A lightweight preprocessing pipeline was adopted to preserve the original linguistic characteristics of the dialectal text. We removed records containing missing values and normalized the sentiment labels to lowercase. No stemming, lemmatization, or manual text normalization was applied, allowing the transformer models to exploit the original lexical and morphological information.

The sentiment labels were mapped to numerical identifiers (*negative*=0, *neutral*=1, and *positive*=2) for model training.

To investigate the impact of explicit dialect information, two input representations were evaluated. In the first configuration, the dialect name was prepended to each sentence (e.g., "[Egyptian] sentence"). In the second configuration, only the original sentence text was used as model input.

4.3 Models

Two transformer-based Arabic language models were evaluated.

MARBERTv2 is a transformer model pretrained on large-scale Arabic social media text with extensive coverage of Arabic dialects. Owing to its dialect-oriented pre-training, it is expected to capture dialect-specific lexical and syntactic variations effectively.

AraBERT is a widely used Arabic pre-trained language model primarily trained on Modern Standard Arabic and diverse Arabic text collections. It serves as a strong baseline for many Arabic NLP tasks and was included to provide a fair comparison against MARBERTv2.

Both models were fine-tuned by replacing the original classification layer with a three-class classification head corresponding to the sentiment categories defined in the shared task.

To ensure a fair comparison, all experiments employed identical preprocessing steps, training settings, and evaluation protocol. The released training data were divided into training and validation sets using an 80/20 stratified split with a fixed random seed of 42, preserving the original class distribution.Models were fine-tuned using the Hugging Face Transformers library. The AdamW optimizer was employed with a learning rate of 2×10^{-5}, a batch size of 16, five training epochs, weight decay of 0.01, and a maximum sequence length of 128 tokens. The best-performing checkpoint was selected according to the validation Macro F1-score.

Three controlled experiments were conducted:

1. Fine-tuning MARBERTv2 using the sentence preceded by its dialect label.
2. Fine-tuning MARBERTv2 using the sentence only.
3. Fine-tuning AraBERT using the sentence only.

The primary evaluation metric was Macro F1-score, following the official shared task evaluation protocol. Overall accuracy, precision, and recall were also reported for comparative analysis.

The implementation was developed in Python 3.12.13 using PyTorch 2.11.0[1], Hugging Face Transformers 5.13.1[2], scikit-learn 1.6.1[3], pandas 2.2.2[4], NumPy 2.0.2[5], and Accelerate 1.14.0[6].

5 Results

5.1 Validation Results

Table 1 summarizes the performance of the three evaluated configurations on the internal validation split.

[1] https://pytorch.org
[2] https://huggingface.co/docs/transformers
[3] https://scikit-learn.org
[4] https://pandas.pydata.org
[5] https://numpy.org
[6] https://github.com/huggingface/accelerate

Table 1: Performance on the internal validation split.

Configuration	Macro F1	Accuracy
MARBERTv2 + Dialect Tag	0.9359	0.9395
MARBERTv2 + Sentence Only	**0.9394**	**0.9424**
AraBERT + Sentence Only	0.9161	0.9222

Three controlled experiments were conducted on the internal validation split while keeping the preprocessing pipeline, training procedure, and evaluation protocol identical.The first experiment evaluated whether explicitly providing dialect information improves sentiment classification. Contrary to our expectation, prepending the dialect label slightly reduced performance. MARBERTv2 achieved a Macro F1-score of 0.9359 when the dialect tag was included, compared with 0.9394 when only the original sentence was used. This suggests that MARBERTv2 is capable of capturing dialect-specific linguistic patterns directly from the sentence without requiring explicit dialect information.The second experiment compared MARBERTv2 with AraBERT under identical experimental settings. MARBERTv2 achieved a higher Macro F1-score (0.9394) than AraBERT (0.9161), indicating that the dialect-oriented pretraining of MARBERTv2 is more suitable for Arabic dialect sentiment classification.At the class level, both models achieved strong performance on the positive and negative classes, whereas the neutral class remained comparatively more challenging. Neutral expressions often share lexical characteristics with both positive and negative instances, making them inherently more difficult to classify.The ablation study demonstrates that removing the explicit dialect tag consistently improves performance, while replacing MARBERTv2 with AraBERT results in a noticeable reduction in Macro F1-score.

5.2 Error Analysis

The best-performing system achieved F1-scores of 0.9695, 0.9005, and 0.9378 for the negative, neutral, and positive classes, respectively. The neutral class obtained the lowest F1-score, indicating that it was the most challenging sentiment category. This is expected because neutral expressions frequently overlap lexically with both positive and negative opinions, increasing the likelihood of ambiguous predictions.

6 Conclusion

This paper presented our system for Subtask 1 of DialectSentEval 2026 on Arabic dialect sentiment classification. We conducted a controlled comparison of MARBERTv2 and AraBERT and further investigated the impact of explicitly incorporating dialect information into the model input.Experimental results on an internal validation split showed that MARBERTv2 achieved the best overall performance, obtaining a Macro F1-score of 0.9394 when using the sentence alone as input. Furthermore, adding explicit dialect labels did not improve classification performance, indicating that the model can effectively learn dialectal characteristics directly from the text.Although the proposed system achieved strong performance, the evaluation was limited to an internal validation split because the official test labels were unavailable during system development.Based on these findings, MARBERTv2 with sentence-only input was selected as our final system for the shared task. Future work includes evaluating the system on the official test set and investigating ensemble learning, data augmentation, and parameter efficient fine tuning to further improve sentiment classification across Arabic dialects.

Acknowledgments

This work was made possible by the National Priorities Research Program grant NPRP14C-0916-210015 from the Qatar National Research Fund (QNRF) part of Qatar Research Development and Innovation Council (QRDI)

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